Markov Chains

Geometric Algorithms

CAS CS 132

Objectives

Finish our discussion of LU factorization

Motivate linear dynamical systems

Analyze Markov chains and their properties

Learn to solve for steady-states

Keywords

linear dynamical systems

recurrence relations

linear difference equations

state vector

probability vector

stochastic matrix

Markov chain

steady-state vector

random walk

state diagram

Recap: LU Factorization

Recall: LU Factorization

Given $A \in \mathbb{R}^{m \times n}$, we'll write A as

echelon form of A

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

Gaussian Elimination and Elementary Matrices

$$A \sim E_1 A \sim E_2 E_1 A \sim \ldots \sim E_k E_{k-1} \ldots E_2 E_1 A$$

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where E implements a *sequence* of row operations

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lower triangle

$$A = E^{-1} B = (E_1^{-1} E_2^{-1} \ldots E_{k-1}^{-1} E_k^{-1}) B$$

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we can do this more efficiently

The forward part of Gaussian elimination is a matrix factorization

demo

How To: LU Factorization by hand

Q. Find a LU Factorization for the matrix A (assuming no swaps)

A.

» Start with L as the identity matrix
» Find U by the forward part of GE
» For each operation $R_i \leftarrow R_i + kR_j$, set L_{ij} to $-k$

Example

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 7 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - 1R_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 1 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Analyzing LU Factorization

Analyzing Linear Algebra Algorithms

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- addition
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$2n$ vs. n is very different when $n \sim 10^{20}$

Dominant Terms

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For polynomials, they're equivalent to their dominant term

Dominant Terms

The dominant term of a polynomial is the monomial with the highest degree

$$\lim_{i \to \infty} \frac{3x^3 + 100000x^2}{3x^3} = 1$$

$3x^3$ dominates the function even though the coefficient for x^2 is so large

How To: Solving systems with the LU

Q. Solve the equation $A\mathbf{x} = \mathbf{b}$ given that $A = LU$ is a LU factorization

A. First solve $L\mathbf{x} = \mathbf{b}$ to get a solution $\mathbf{c}$, then solve $U\mathbf{x} = \mathbf{c}$ to get a solution $\mathbf{d}$

$L\vec{r} = \vec{b}$ $\quad U\vec{w} = \vec{r}$

$A\vec{w} = LU\vec{w}$

$= L\vec{r}$

$= \vec{b}$

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Why is this better than just solving $A\mathbf{x} = \mathbf{b}$***?***

FLOPs for Solving General Systems

The following FLOP estimates are based on $n \times n$ matrices

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Solving by LU Factorization: $\sim \frac{2n^3}{3}$ FLOPS

If you solve several matrix equations for the same matrix, LU factorization is faster than matrix inversion on the first equation, and the same (asymptotically) in later equations (and it works for rectangular matrices).

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If A doesn't have to many nonzero entries (A is **sparse)** then its likely that L and U won't either.

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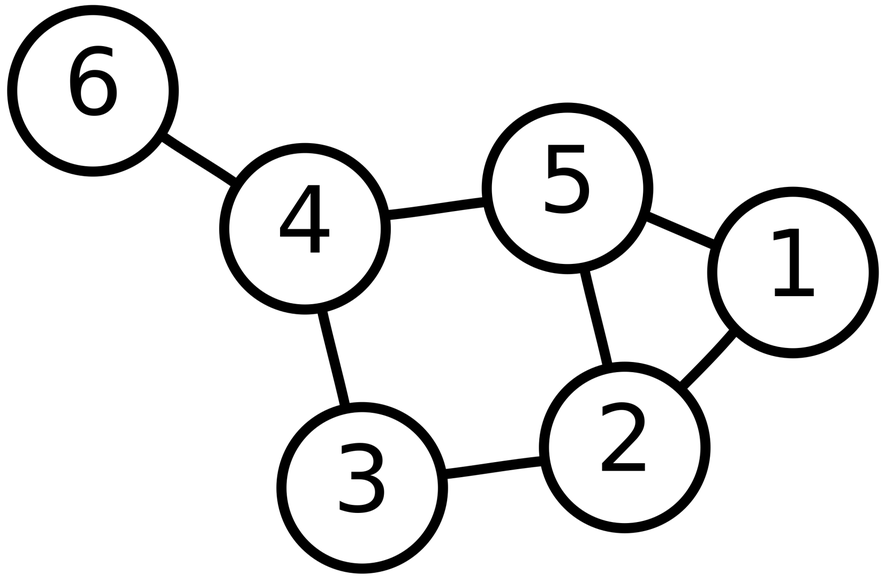
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Sparse matrices are faster to compute with and better with respect to storage

Graph Theory Primer

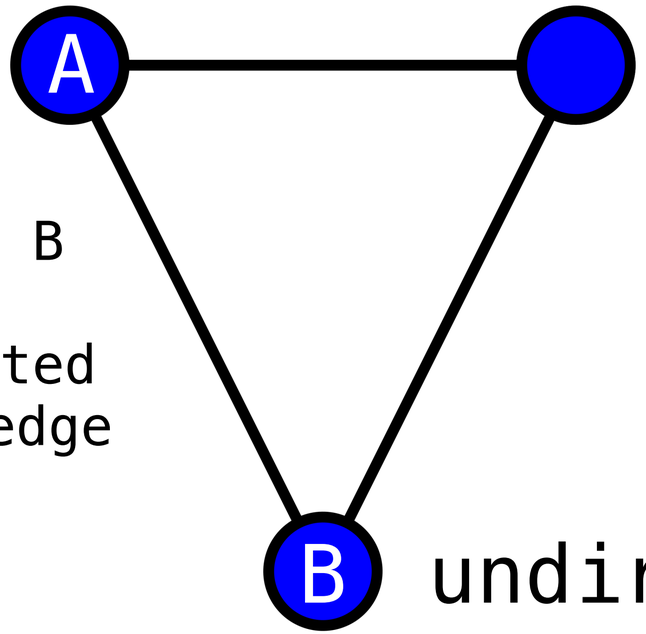
Graphs

Def (Informal). A **graph** is a collection of nodes with edges between them



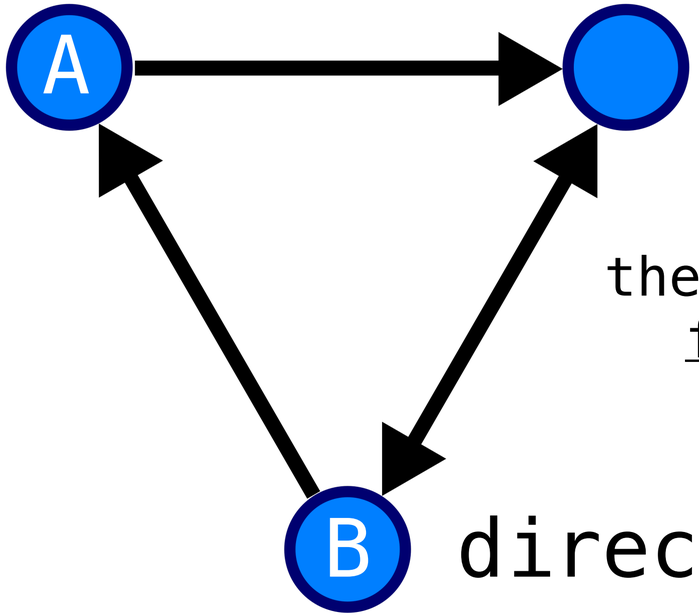
Directed vs. Undirected Graphs

A graph is **directed** if its edges have a direction



A and B are connected by an edge

undirected

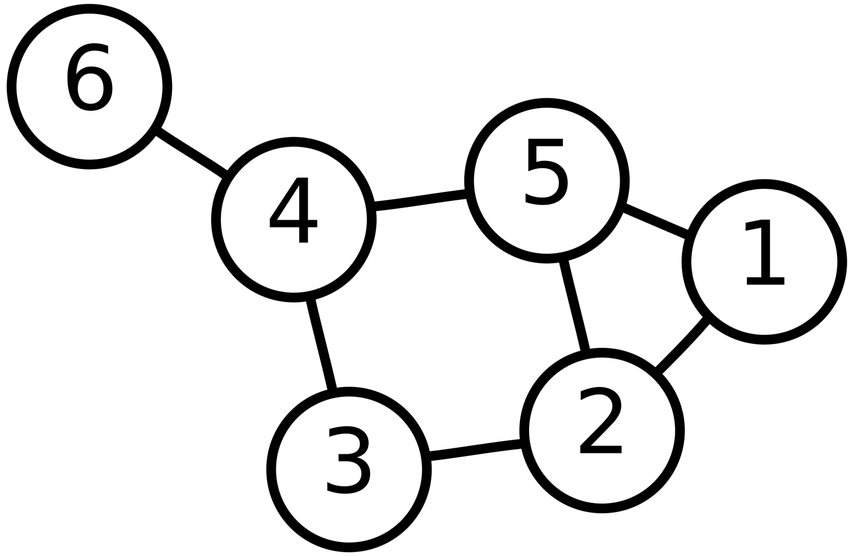


there is an edge from B to A

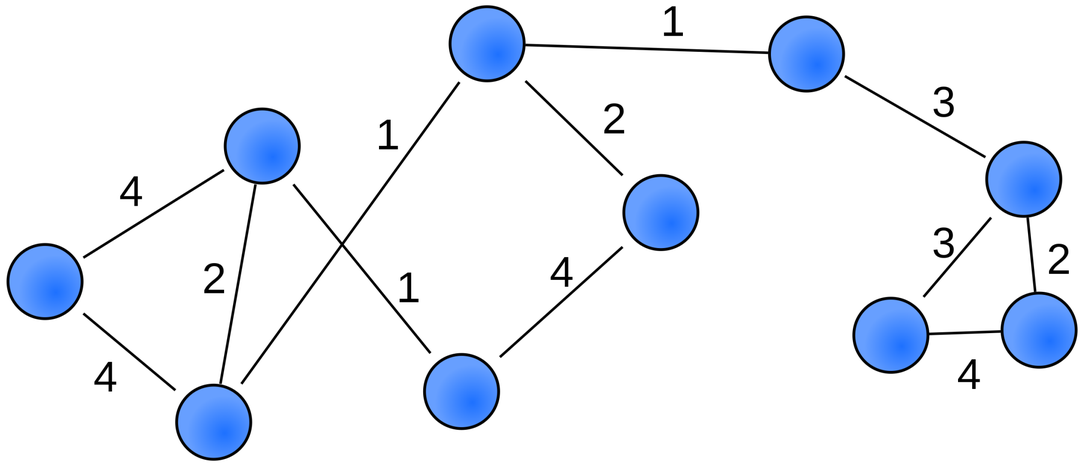
directed

Weighted vs Unweighted graphs

A graph is **weighted** if its edges have associated values



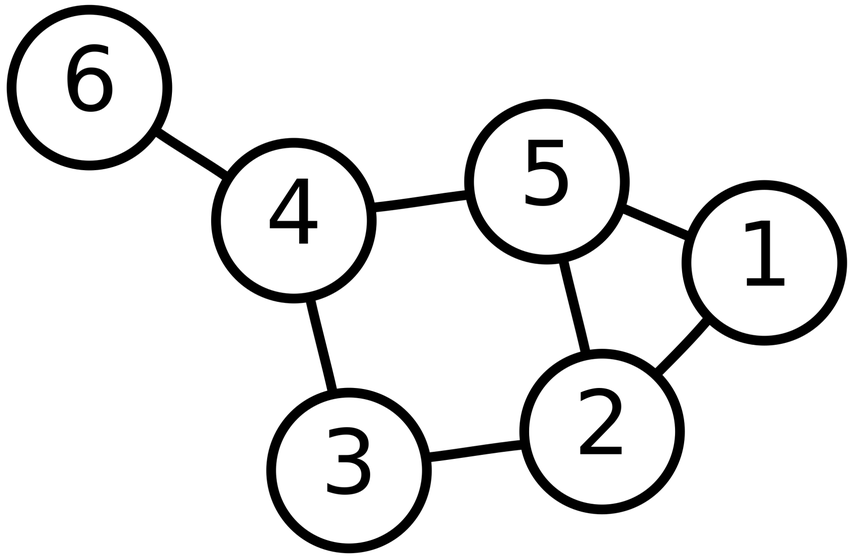
unweighted



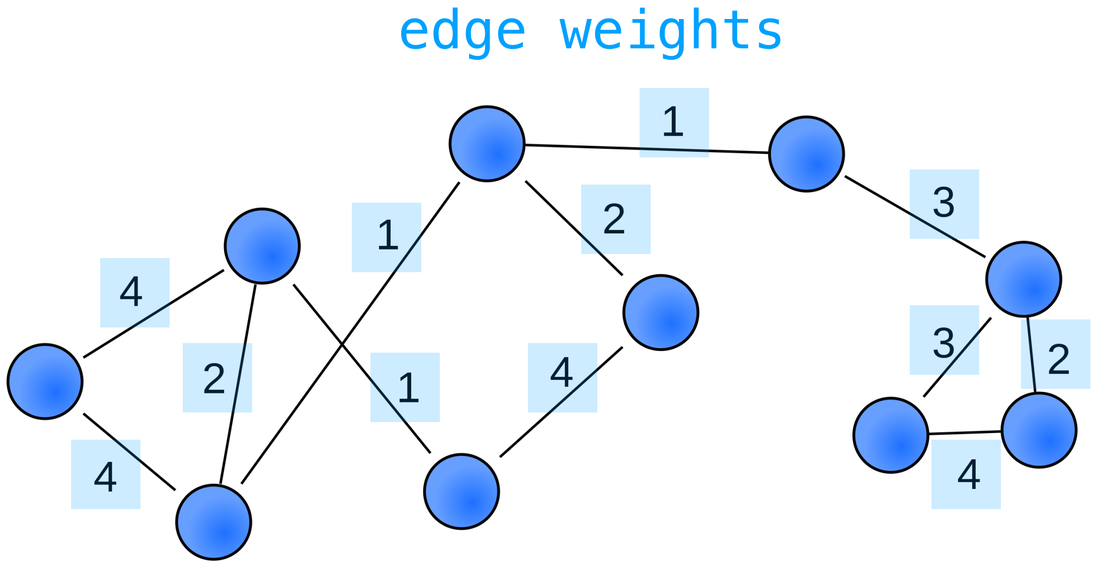
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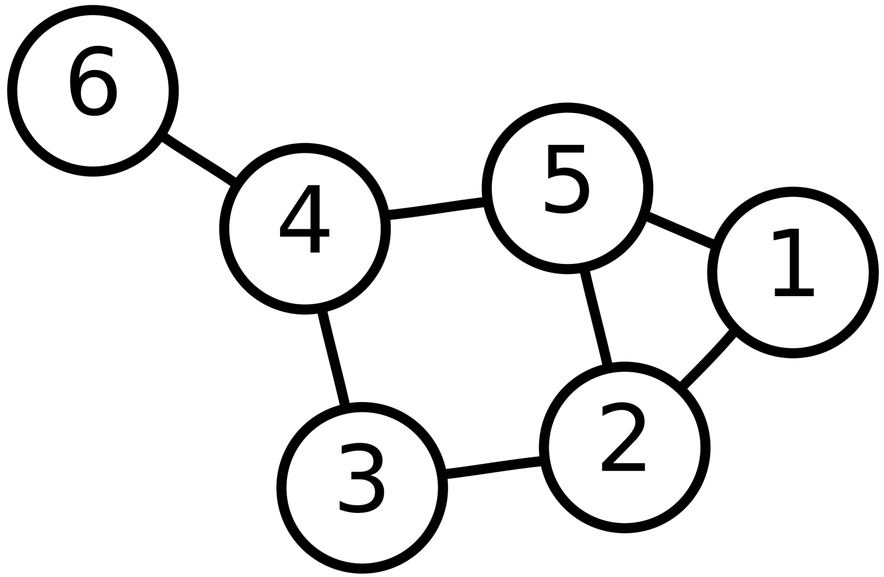
unweighted



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Simple Graphs

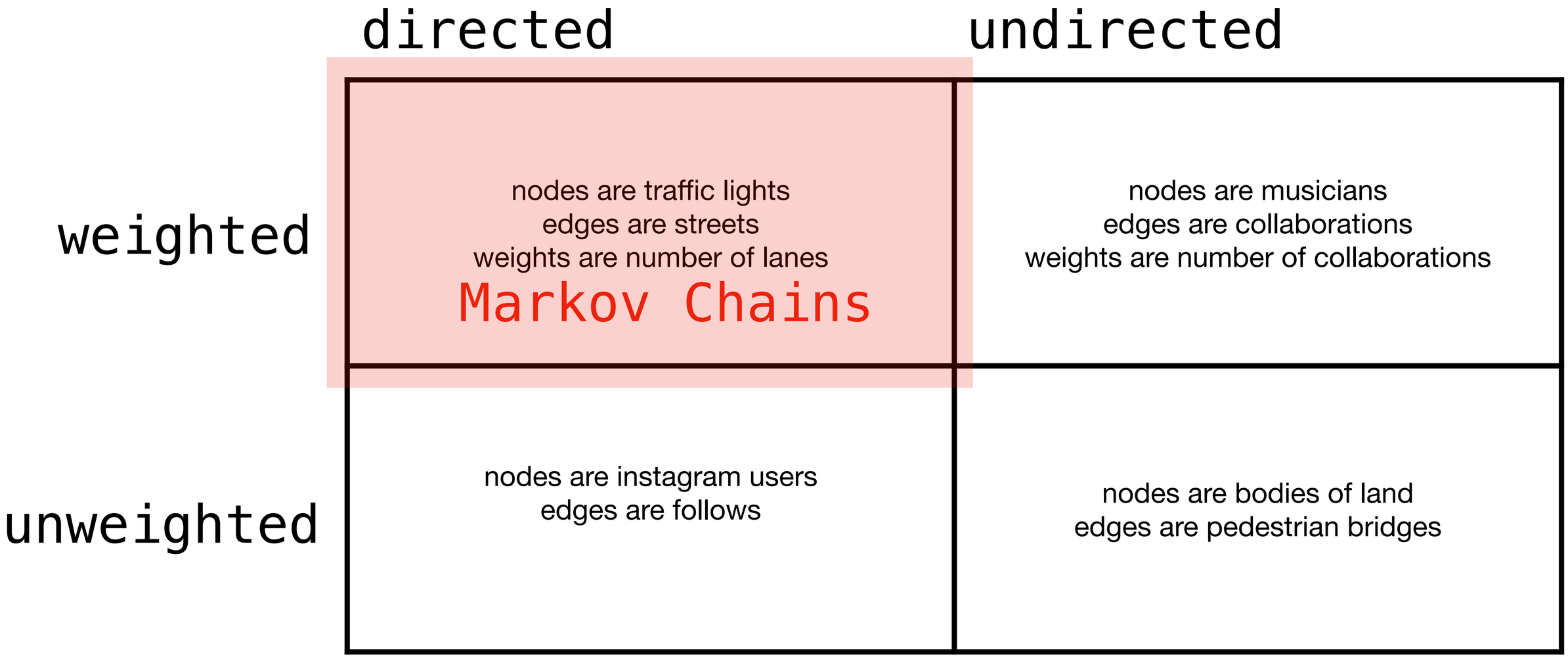
A graph is **simple** if it is undirected, has no self loops, and no multi-edges



Four Kinds of Graphs

	directed	undirected
weighted	nodes are traffic lights edges are streets weights are number of lanes	nodes are musicians edges are collaborations weights are number of collaborations
unweighted	nodes are instagram users edges are follows	nodes are bodies of land edges are pedestrian bridges

Four Kinds of Graphs



Fundamental Question

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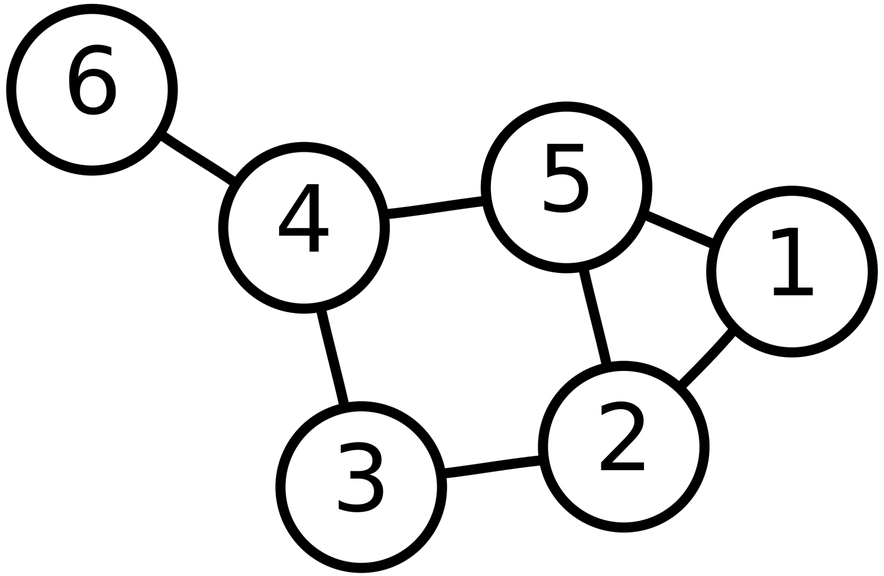
How do we represent a graph formally in a computer?

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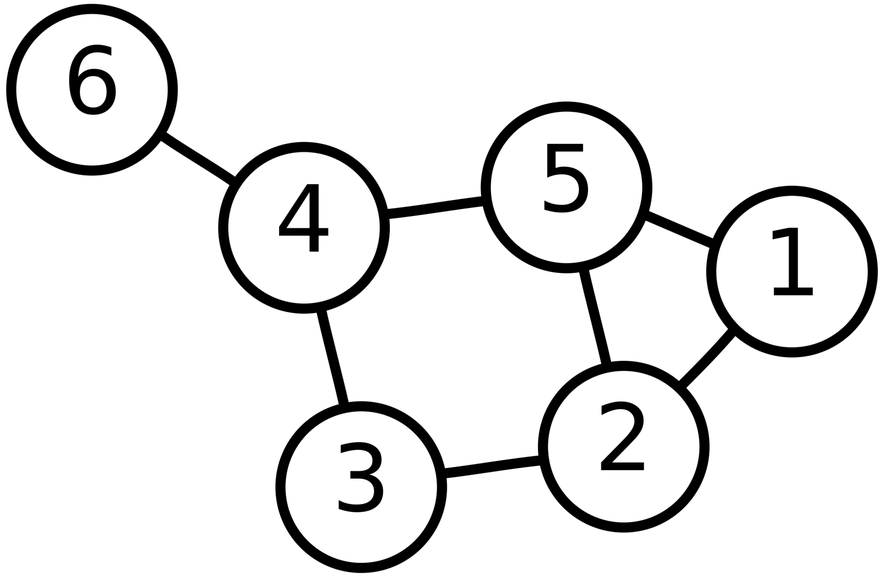
There are a couple ways, but one way is to use **matrices**

Adjacency Matrices



Adjacency Matrices

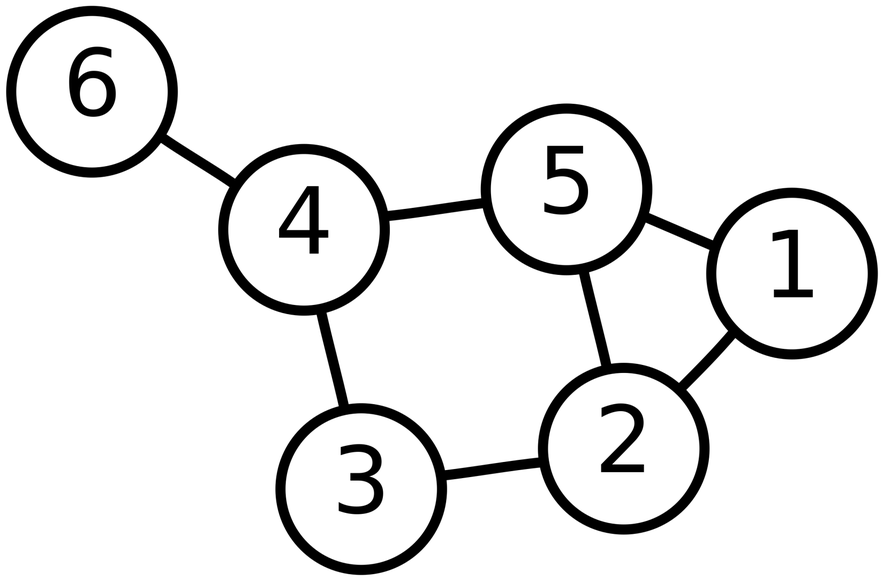
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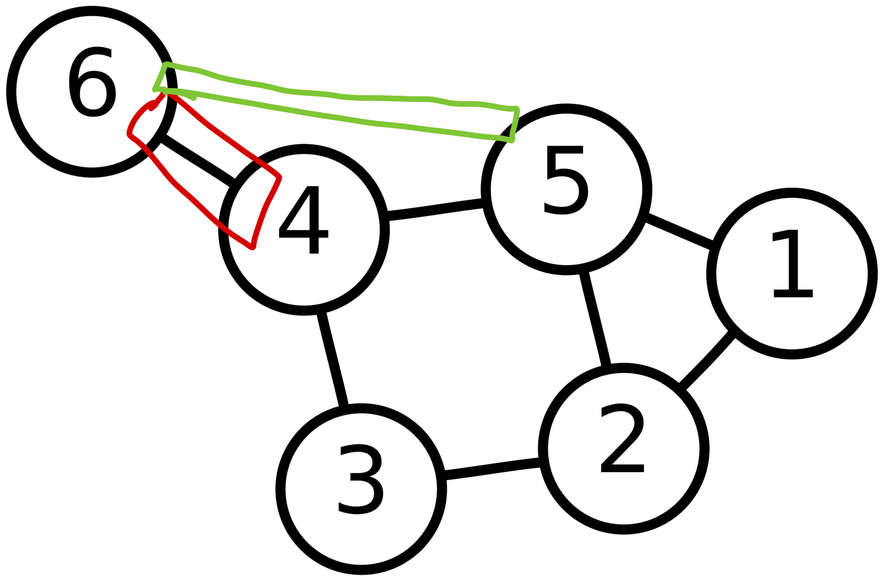
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$$A_{ij} = \begin{cases} 1 & \text{edge between i and j} \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$



Once we have adjacency matrices we can do linear algebra on graphs

Dynamical Systems

Change

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Things change from one state of affairs to another state of affairs

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Things change from one state of affairs to another state of affairs

Things change often in unpredictable ways

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If something changes unpredictably, what can we say about it?

Dynamical Systems

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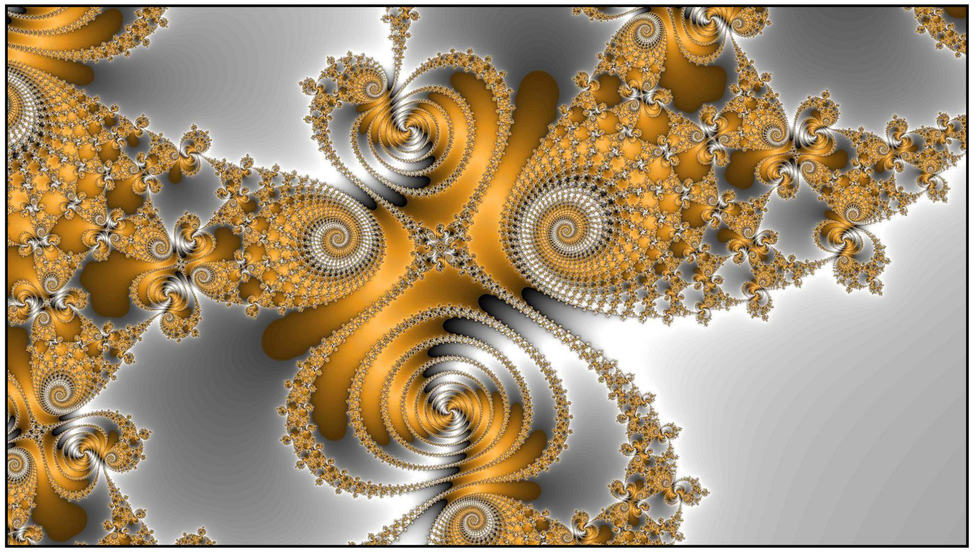
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» economics (stocks)
» physical/chemical systems
» populations
» weather

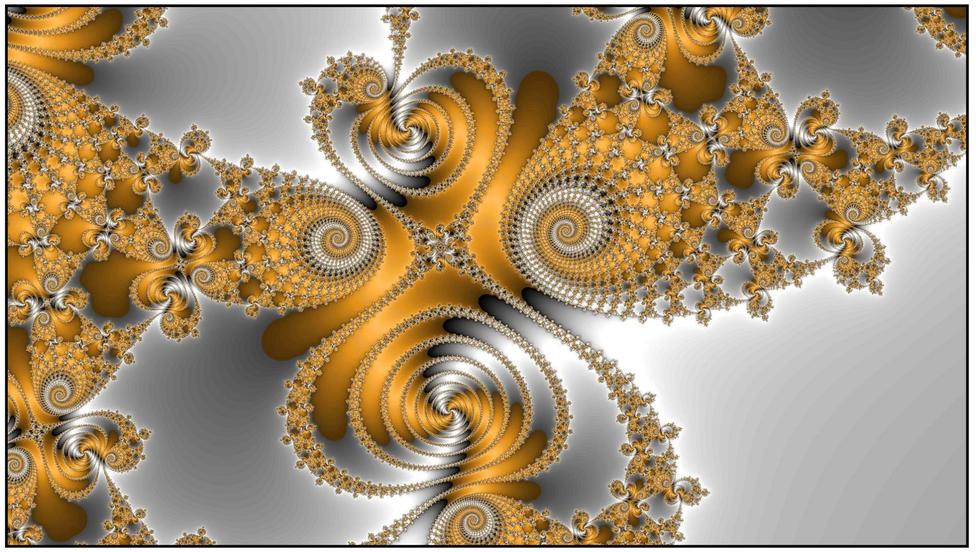
An Aside: Chaos Theory



https://commons.wikimedia.org/wiki/File:Fr137.jpg

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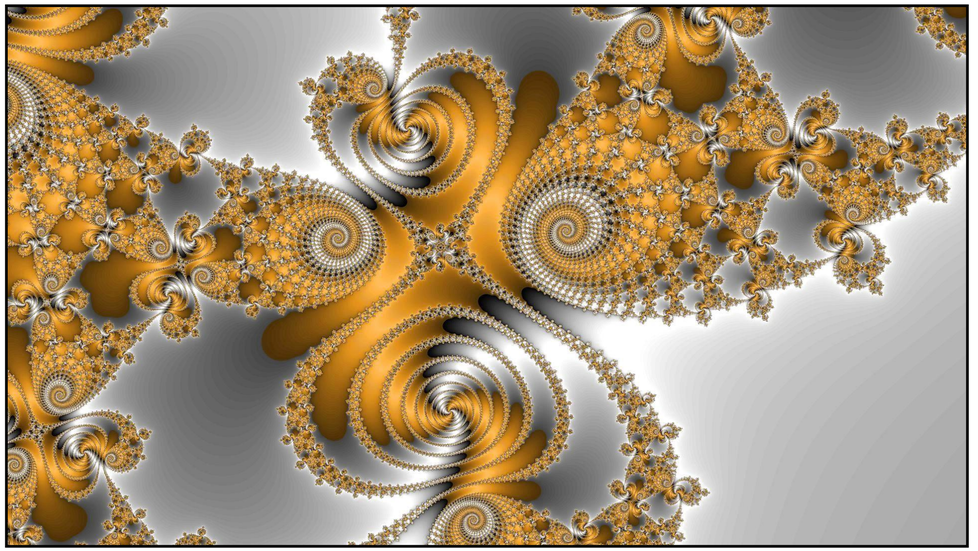


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But even in chaotic systems
there are *underlying patterns*
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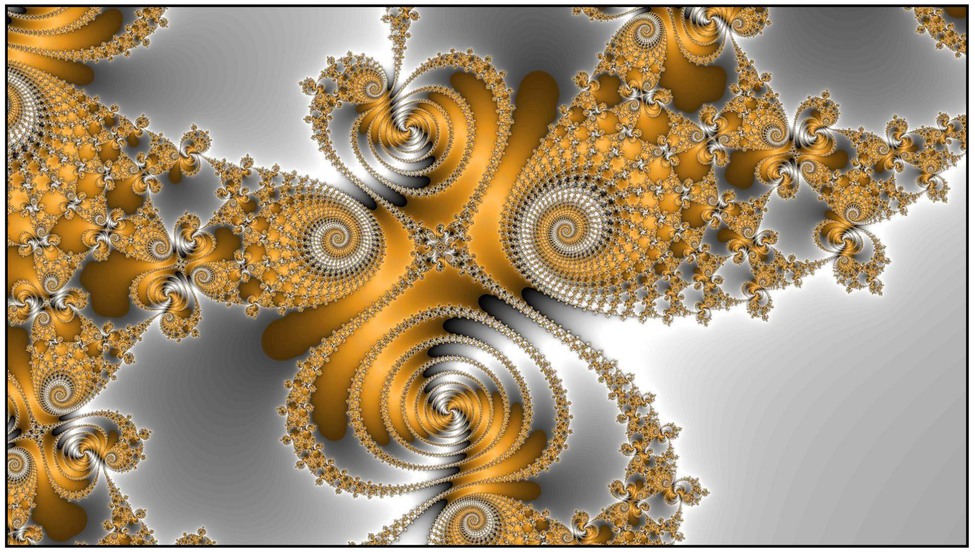
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An Aside: Chaos Theory

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But even in chaotic systems there are *underlying patterns* and *repeated structures*

Often it's useful to consider chaotic systems in terms of *global* properties



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Does a part of the system *dominate* over time?

Linear Dynamical Systems

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Given an **initial state vector** $\mathbf{v}_0$**,** we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1}$$ *(linear recurrence relation)*

State Vectors

$$\vec{v}_k = A^k \vec{v}_0$$

$$
\begin{aligned}
&\mathbf{v}_0 \\
&\mathbf{v}_1 = A\mathbf{v}_0 \\
&\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0) \\
&\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0) \\
&\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0) \\
&\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0) \\
&\quad\vdots
\end{aligned}
$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

Warm up:
Population Dynamics

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We find by analyzing previous data that each year:

- 5% of the population moves from city → suburb
- 3% of the population moves from suburb → city

Fundamental Question

Can we make any predictions about the population of the city and suburb in 20 years?

Assumptions: No immigration, emigration, birth, death, etc. **The overall population stays fixed.**

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people who stayed

people who left

Setting up a Matrix

$$\begin{bmatrix} \text{city}_1 \\ \text{suburb}_1 \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix} = \begin{bmatrix} 582{,}000 \\ 418{,}000 \end{bmatrix}$$

We expect the population of the city to decrease in a year

Setting up a Matrix

$$\begin{bmatrix} \text{city}_2 \\ \text{suburb}_2 \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_1 \\ \text{suburb}_1 \end{bmatrix} = \begin{bmatrix} 565{,}440 \\ 434{,}560 \end{bmatrix}$$

The next year, we expect the population of the city to *continue* to decrease

Will it decrease indefinitely?

Setting up a Matrix

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_{k-1} \\ \text{suburb}_{k-1} \end{bmatrix}$$

Setting up a Matrix

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_{k-1} \\ \text{suburb}_{k-1} \end{bmatrix}$$

This is a *linear dynamical system*

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This is a *linear dynamical system*

To guess what the population will look like in 20 years, we need to compute

$$\begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}^{20} \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix}$$

demo

Markov Chains

Stochastic Matrices

$$\begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$$

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What's special about this matrix?

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- Its columns sum to 1

This should make us think probability

Stochastic Matrices

Def. A matrix in $\mathbb{R}^{n \times n}$ is **stochastic** if its entries are nonnegative and its columns sum to 1

$$\begin{bmatrix} 0.7 & 0.1 & 0.3 \\ 0.2 & 0.8 & 0.3 \\ 0.1 & 0.1 & 0.4 \end{bmatrix}$$

Markov Chains

Def. A **Markov chain** is a linear dynamical system whose evolution function is given by a *stochastic* matrix

State Diagrams

$$\begin{array}{c} \\ \text{city} \\ \\ \text{suburb} \end{array} \begin{array}{c} \begin{array}{cc} \text{city} & \text{sub.} \end{array} \\ \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix} \end{array}$$

Def. A **state diagram** is a directed weighted graph whose adjacency matrix is stochastic.

Ex.

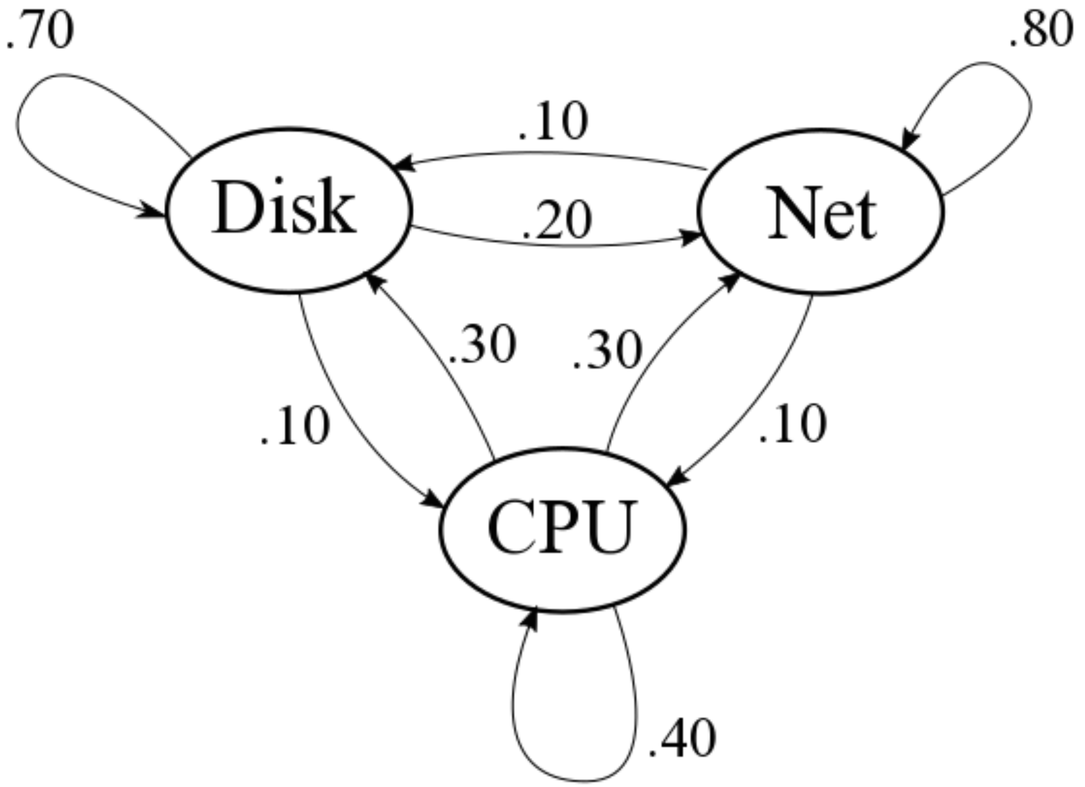


Another Example: Computer System

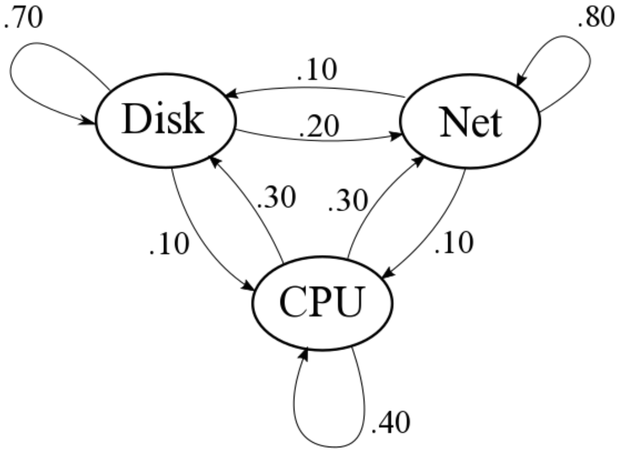
Imagine a computer system in which tasks request service from disk, network or CPU

In the long term, which device is busiest?

This is about finding a stable state



Example



Key Property of Stochastic Matrices

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Stochastic matrices *redistribute* the "stuff" in a vector

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Thm. For a stochastic matrix A and a vector $\mathbf{v}$,

sum of entries of $\mathbf{v}$ = sum of entries of $A\mathbf{v}$

Key Property of Stochastic Matrices

Thm. For stochastic A and a vector $\mathbf{v}$

$$\mathbf{1}^T\mathbf{v} = \mathbf{1}^T(A\mathbf{v})$$

Exercise: show this holds

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_{k-1} \\ \text{suburb}_{k-1} \end{bmatrix}$$

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix}$$

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} 600{,}000 \\ 400{,}000 \end{bmatrix}$$

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} 600{,}000 \\ 400{,}000 \end{bmatrix}$$

What if we want to analyze a different population?

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

60% of pop. in city

40% of pop. in suburb

We should think about **how the population is distributed** instead of a particular population

Probability Vectors

Probability Vectors

Def. A **probability vector** is a vector of nonnegative values that sum to 1. They represent:

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Def. A **probability vector** is a vector of nonnegative values that sum to 1. They represent:

» discrete probability distributions

» distributions of collections of things

(these are really the same thing)

Probability Vectors (Example)

The vector $\begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$ represented the distribution of the population, but we can also think of this as:

If we choose a random person from the population we'll get someone:

in the city with probability 0.6

in the suburbs with probability 0.4

The Point

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We'll be interested in the dynamics of Markov chains on **probability vectors**

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Since stochastic matrices preserve $\mathbf{1}^T\mathbf{v}$, they *transform* one distribution into another

Can we say something about how the distribution changes in the long run?

Steady-State Vectors

Steady-State Vectors

Def. A **steady-state vector** for a stochastic matrix A is a probability vector $\mathbf{q}$ such that

$$A\mathbf{q} = \mathbf{q}$$

A steady-state vector is *not changed* by the stochastic matrix. They describe *equilibriums*

Returning to the Example

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How do we interpret a steady-state vector for our example?

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The populations in the city and the suburb stay the same over time

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How do we interpret a steady-state vector for our example?

The populations in the city and the suburb stay the same over time

The same number of people are moving into and out of the city each year

Fundamental Questions

Do steady states exist?

Are they unique?

How do we find them?

Finding Steady-State Vectors

$$A\mathbf{q} = I\mathbf{q}$$

Let's solve this equation for $\mathbf{q}$:

$$A\vec{q} - I\vec{q} = \vec{0}$$

$$(A - I)\vec{q} = \vec{0}$$

homogen. matrix eq.

How to: Steady-State Vectors

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Q. Determine if the Markov chain with stochastic matrix A has a steady-state vector. If it does, find such a vector

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How to: Steady-State Vectors

Q. Determine if the Markov chain with stochastic matrix A has a steady-state vector. If it does, find such a vector

A. Solve the equation $(A - I)\mathbf{x} = \mathbf{0}$ and find a solution whose entries sum to 1 (this will be possible given a free variable)

If there is no such solution, the system does not have a steady state

Example

$$A = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$$

$$A - I = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.05 & 0.03 \\ 0.05 & -0.03 \end{bmatrix} \sim \begin{bmatrix} -5 & 3 \\ 5 & -3 \end{bmatrix} \sim \begin{bmatrix} -5 & 3 \\ 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -3/5 \\ 0 & 0 \end{bmatrix}$$

$x_1 = 3/5\, x_2$

x_2 is free

$x_1 + x_2 = 1$

$\Downarrow$

$\frac{3}{5} x_2 + x_2 = 1$

$\Downarrow$

$\frac{8}{5} x_2 = 1$

$x_1 = 3/8$ $\quad x = 5/8$

$$\begin{bmatrix} 3/8 \\ 5/8 \end{bmatrix}$$

$$A \quad \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}$$

demo

Existence vs Convergence

If $(A - I)\mathbf{x} = \mathbf{0}$ has a nontrival solutions, then the system has a stable state. This **does not** mean:

- the stable state is unique
- the system *converges* to this state

Convergence

Convergence

Def. For a Markov chain with stochastic matrix A, an initial state $\mathbf{v}_0$ **converges** to the state $\mathbf{v}$ if $\lim_{k\to\infty} A^k\mathbf{v}_0 = \mathbf{v}$

Convergence

Def. For a Markov chain with stochastic matrix A, an initial state $\mathbf{v}_0$ **converges** to the state $\mathbf{v}$ if $\lim_{k\to\infty} A^k\mathbf{v}_0 = \mathbf{v}$

As we repeatedly multiply $\mathbf{v}_0$ by A, we get closer and closer to $\mathbf{v}$ (in the limit)

Example of Non-Convergence

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Non-Example. I is a stochastic matrix and

$$I\mathbf{v} = \mathbf{v}$$

for any choice of $\mathbf{v}$

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$$I\mathbf{v} = \mathbf{v}$$

for any choice of $\mathbf{v}$

So this system does not have a unique steady state

And no vectors converge to the same stable state

Regular Stochastic Matrices

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Definition. A stochastic matrix A is **regular** if A^k has all positive entries for *some nonnegative* k

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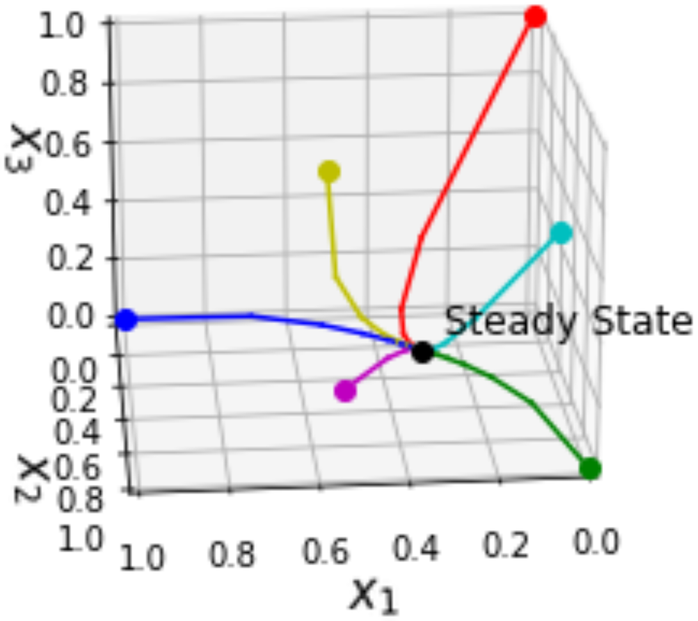
Theorem. A regular stochastic matrix P has a unique steady state, and

every probability vector converges to it

Mixing

This process of converging to a unique steady state is called "mixing"

This theorem says, after some amount of mixing, we'll be close to the stable state **no matter where we started**



Example

$$\begin{bmatrix} 0.5 & 0.4 & 0 \\ 0.5 & 0.4 & 0.5 \\ 0 & 0.2 & 0.5 \end{bmatrix}$$

Summary

Markov chains allow us to reason about dynamical systems that are dictated by some amount of randomness

Stable states represent equilibriums